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Roll no.

Degree: B.Tech. (ECAM) Semester: 3rd Course work
END-SEMESTER THEORY EXAMINATION, NOV-DEC,2024

Course Code: EAEPC303 / EAEPC06

Course Title: Probability Theory and Random Process

Time: 03 Hours

Max. Marks: 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	The number of customers arriving at a store follows a Poisson distribution with a mean of 4 customers per hour. Find: i) $P(X=3)$ where X is the number of customers arriving in one hour and ii) $P(X \geq 5)$	5	CO3
1b	Derive the expression of characteristic function for Gaussian distribution	5	CO4
1c	i) A random variable X follows a Poisson distribution with parameter $\lambda=3$. Find the probability $P(X=2)$. ii) Let X be a Binomial random variable with $n=5$ trials and $p=0.4$ success probability. Find the probability $P(X=3)$.	5	CO3
Q 2	Attempt any 2 parts of the following		
2a	Explain joint distribution function and joint density functions. If X and Y are independent random variables, then prove that, $E[Y/X] = E(Y)$.	5	CO2,5
2b	Write down the expression of covariance of X and Y and explain its significance. If X and Y are uncorrelated random variables, then prove that $\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2$	5	CO1, 2, 5
2c	Consider two random variables X and Y with the following properties: The covariance between X and Y is 0; and the correlation coefficient $\rho(X,Y)=0.8$. Are X and Y independent based on the given covariance and correlation? Explain why or why not.	5	CO2, 5

Q 3	Attempt any 2 parts of the following		
3a	Let X be a random variable with $X \sim N(2,1)$. Find the distribution of $Y = X^2$	5	CO4
3b	Explain and prove Chebyshev's Inequality.	5	CO2, 5
3c	Consider a random variable X , with the PDF $f_X(x)$. Derive the PDF of Y where $Y = g(X)$.	5	CO4
Q 4	Attempt any 2 parts of the following		
4a	Write down and explain the properties of time autocorrelation function.	5	CO1,2
4b	Let $x(t) = a \cos(wt + \alpha)$ is a Random process, w is a constant and a and α are independent RVs uniformly distributed in the ranges $(-1, 1)$ and $(0, 2\pi)$, respectively. (i) Sketch the ensemble of this process. (ii) Determine time average of $x(t)$. (iii) Determine $R_{xx}(t_1, t_2)$. (iv) Is the process wide-sense stationary? (v) Is the process ergodic?	5	CO3, 4
4c	The autocorrelation function of a stationary process is given by, $R_{xx}(\tau) = 25 + \frac{4}{1 + 6\tau^2}$. Find mean value and variance of the process.	5	CO3
Q 5	Attempt any 2 parts of the following		
5a	Consider a linear time-invariant system with an input random process $X(t)$ and output $Y(t)$ i) Derive the autocorrelation function of the output $Y(t)$ in terms of the input $X(t)$ and the system's impulse response. ii) Explain how the power spectral density of the output is related to the input power spectral density for an LTI system.	5	CO 2, 4
5b	Derive the relationship between the power density spectrum and the autocorrelation function of random process.	5	CO 2, 4
5c	What is "white noise"? Derive the autocorrelation function of white noise. Why is white noise unrealizable?	5	CO2, 4